Preference Optimization

Andrei Filatov · 2026
t.me/awesome_dl · anvilarth.github.io

Talk Roadmap

1. Direct Preference Optimization (DPO)
2. FiFA - automatic DPO
3. Manifold drift
    a. Rethinking DPO
    b. ThermoDPO

Preference Optimization

Pretraining is about {modality1}-{modality2} alignment. But it's lacking visual prioritization (e.g. this lighting is better, this texture is worse, etc.)





Preference Optimization

What type of preference optimization do we have?

1. **Pair-based – DPO**. Explicit feedback model/person – this one is better / this one is worse

2. **Reward based – RL, GRPO**, etc. Reward model provides granular feedback (OCR, Aesthetics)

Direct Preference Optimization

In LLMs, preference optimization started with RLHF/DPO

$$L_{\text{DPO}}(\theta) = -\mathbb{E}_{\boldsymbol{c},\boldsymbol{x}_0^w,\boldsymbol{x}_0^l}\left[\log\sigma\left(\beta\log\frac{p_\theta(\boldsymbol{x}_0^w|\boldsymbol{c})}{p_{\text{ref}}(\boldsymbol{x}_0^w|\boldsymbol{c})} - \beta\log\frac{p_\theta(\boldsymbol{x}_0^l|\boldsymbol{c})}{p_{\text{ref}}(\boldsymbol{x}_0^l|\boldsymbol{c})}\right)\right] \quad (8)$$

How to adapt it to diffusion models?

Direct Preference Optimization

$$L_{\text{DPO}}(\theta) = -\mathbb{E}_{\boldsymbol{c}, \boldsymbol{x}_0^w, \boldsymbol{x}_0^l} \left[\log \sigma \left(\beta \log \frac{p_\theta(\boldsymbol{x}_0^w | \boldsymbol{c})}{p_{\text{ref}}(\boldsymbol{x}_0^w | \boldsymbol{c})} - \beta \log \frac{p_\theta(\boldsymbol{x}_0^l | \boldsymbol{c})}{p_{\text{ref}}(\boldsymbol{x}_0^l | \boldsymbol{c})} \right) \right]$$

$$L_{\text{DPO-Diffusion}}(\theta) = -\mathbb{E}_{(\boldsymbol{x}_0^w, \boldsymbol{x}_0^l) \sim \mathcal{D}} \log \sigma \Bigg(\beta \mathbb{E}_{\substack{\boldsymbol{x}_{1:T}^w \sim p_\theta(\boldsymbol{x}_{1:T}^w | \boldsymbol{x}_0^w) \\ \boldsymbol{x}_{1:T}^l \sim p_\theta(\boldsymbol{x}_{1:T}^l | \boldsymbol{x}_0^l)}} \left[\log \frac{p_\theta(\boldsymbol{x}_{0:T}^w)}{p_{\text{ref}}(\boldsymbol{x}_{0:T}^w)} - \log \frac{p_\theta(\boldsymbol{x}_{0:T}^l)}{p_{\text{ref}}(\boldsymbol{x}_{0:T}^l)} \right] \Bigg)$$

Direct Preference Optimization

If we do some ELBO / Jensen Inequality math, we get this final form

$$\log \sigma \left(\beta T \log \frac{p_\theta(\boldsymbol{x}_{t-1}^w | \boldsymbol{x}_t^w)}{p_{\text{ref}}(\boldsymbol{x}_{t-1}^w | \boldsymbol{x}_t^w)} - \beta T \log \frac{p_\theta(\boldsymbol{x}_{t-1}^l | \boldsymbol{x}_t^l)}{p_{\text{ref}}(\boldsymbol{x}_{t-1}^l | \boldsymbol{x}_t^l)} \right)$$

Direct Preference Optimization

Final loss:

$$
\begin{aligned}
L(\theta) = -\mathbb{E}_{(\boldsymbol{x}_0^w, \boldsymbol{x}_0^l) \sim \mathcal{D}, t \sim \mathcal{U}(0,T), \boldsymbol{x}_t^w \sim q(\boldsymbol{x}_t^w | \boldsymbol{x}_0^w), \boldsymbol{x}_t^l \sim q(\boldsymbol{x}_t^l | \boldsymbol{x}_0^l)} \\
\log \sigma \left(-\beta T \omega(\lambda_t) \right(\\
\| \boldsymbol{\epsilon}^w - \boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t^w, t) \|_2^2 - \| \boldsymbol{\epsilon}^w - \boldsymbol{\epsilon}_{\text{ref}}(\boldsymbol{x}_t^w, t) \|_2^2 \\
- \left(\| \boldsymbol{\epsilon}^l - \boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t^l, t) \|_2^2 - \| \boldsymbol{\epsilon}^l - \boldsymbol{\epsilon}_{\text{ref}}(\boldsymbol{x}_t^l, t) \|_2^2 \right) \right) \right)
\end{aligned} \tag{14}
$$

Results (DPO on Pick-a-Pic)



Ablation of beta

Beta doesn’t affect final result but it affects training pace.

$$\frac{2000}{\beta} 2.048 \cdot 10^{-8}$$

Learning rate

Higher beta -> faster saturation, low controllability



Direct Preference Optimization

Trained model can be used for assessing margins

Model	PS	HPS	CLIP	Aes.	DPO-SD1.5	DPO-SDXL
Acc.	64.2	59.3	57.1	51.4	60.8	72.0

Direct Preference Optimization

1) First round is usually synthetic/automatic DPO. As you can squeeze a lot w/o human annotation.

2) Human annotation round.
   a) Create rollouts with dpo_v1 model
   b) Assess with people
   c) Train another round

DPO problems

- Pair choice:

  On-policy rollouts usually produce small margins. To improve it, we use prompt expander + data filtering -> FiFA.

- Domain drift - loser component force model to leave distribution.

  -> Rethinking DPO, ThermoDPO

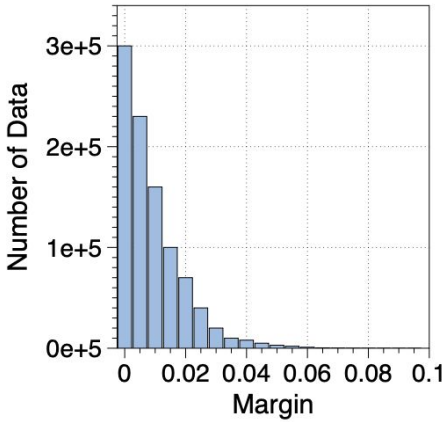
- It’s hard to control improvement but we can improve barely described things (RL is your next choice)

FiFA

- Preference data is noisy (biased annotators + biased model) -> weak signal -> preference is hard to learn



(a) Samples of high/low preference margin



(b) Distribution of reward margin

FiFA

Let’s use high margin data in training:

- Text diversity - prompts in training are diverse (Entropy)
- Text quality - prompts in training are good (gpt-score)
- Preference margin - Train model for preference (Clip-based)

Algorithm 1: Algorithm for **FiFA**

Input: Initial dataset $D = \{\mathbf{c}_i, \mathbf{x}^w_{0,i}, \mathbf{x}^l_{0,i}\}_{i=1}^N$, LLM model for scoring $LLM_Score(\cdot)$, Reward model $r_\phi(\cdot, \cdot)$, Hyperparameters for quality α and diversity γ, Number of filtered data points K

Output: Filtered dataset $S = \{\mathbf{c}_i, \mathbf{x}^w_{0,i}, \mathbf{x}^l_{0,i}\}_{i=1}^K$

$S \leftarrow \{\}$ `// Initialize the filtered dataset as empty`

for each data point $(\mathbf{c}_i, \mathbf{x}^w_{0,i}, \mathbf{x}^l_{0,i})$ in D **do**

  $m_i^{reward} \leftarrow |r_\phi(\mathbf{c}_i, \mathbf{x}^w_{0,i}) - r_\phi(\mathbf{c}_i, \mathbf{x}^l_{0,i})|$ `// Calculate the reward margin for each data point`

  $\tilde{f}(\mathbf{c}_i, \mathbf{x}^w_{0,i}, \mathbf{x}^l_{0,i}) \leftarrow m_i^{reward} + \alpha * LLM_Score(\mathbf{c}_i) + \gamma * \log \|\mathbf{c}_i - \mathbf{c}_i^{k\text{-}NN}\|_2$

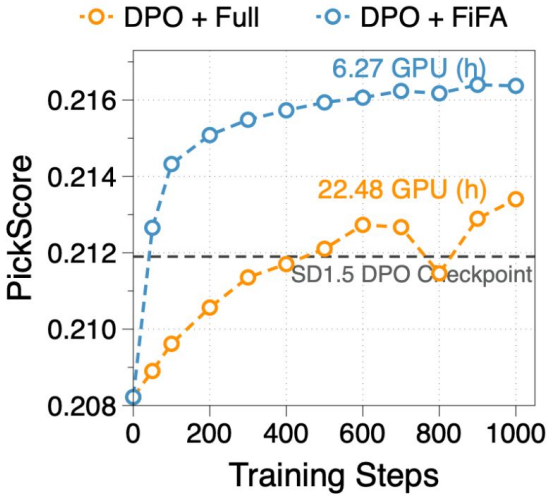
  `// Compute the data importance score` $\tilde{f}$ `for each data point`

end for

Sort data points in D by $\tilde{f}(\mathbf{c}_i, \mathbf{x}^w_{0,i}, \mathbf{x}^l_{0,i})$ in descending order.

Select the top K data points based on $\tilde{f}$ to form S.

return S



(a) PickScore by each training step



(b) Qualitative examples of generated images

Pretrain | DPO + Full | DPO + FiFA



"A **pineapple bean bag** designed by Vladimir Kush and Ilya Kuvshinov"

Pretrain | DPO + Full | DPO + FiFA



"**Superman** with Danny DeVito's **face**"



"A head and shoulders portrait of a **black cartoon rabbit** wearing a shirt and laughing with big eyes in the style of Walt Disney animation"



"**A girl** gazes at a city from a mountain at night in a colored manga illustration by Diego Facio"

FiFA

- Main result: quality >>> quantity in DPO (read Emu)
- You can induce scores by choosing high margin pairs (chatgpt vs qwen, hello qwen)

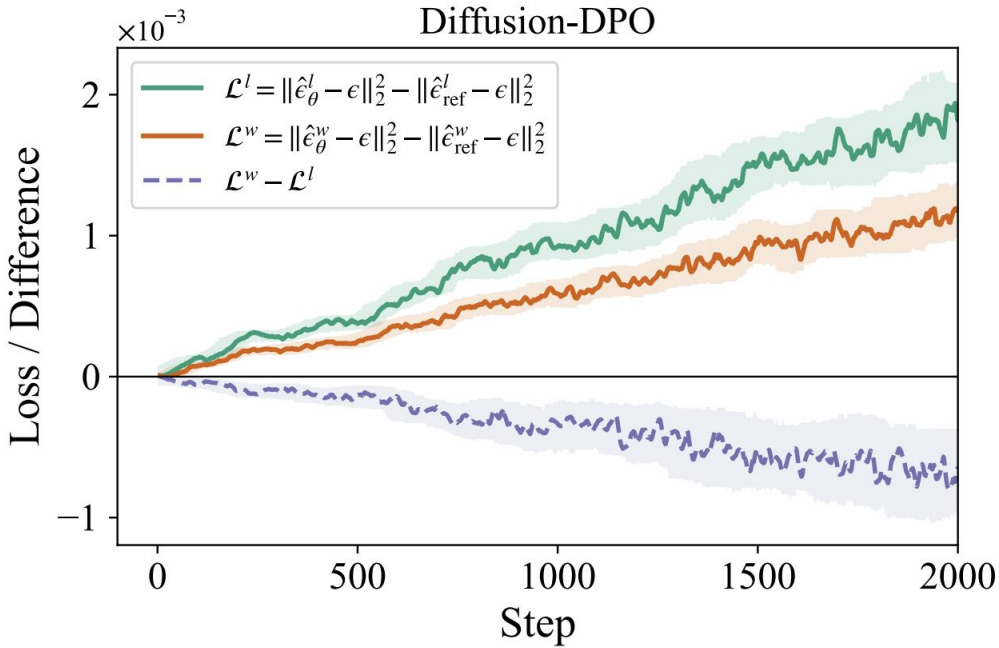
Rethinking DPO

- When you are doing DPO for model it’s easy to forget loser distribution to minimize loss:

    - It’s good for removing artifacts but if two pictures almost the same you can fall in biases
    - Common practice is to add SFT regularization

$$\mathcal{L}_{\text{DPO+NLL}} = \mathcal{L}_{\text{DPO}} + \alpha \mathcal{L}_{\text{NLL}},$$

Diffusion DSPO



Rethinking DPO

- Here authors propose to limit loss by some mixtures – which is balance between DPO and SFT on winner. If the loser term is too small, fall back to sigmoid SFT on the winner.

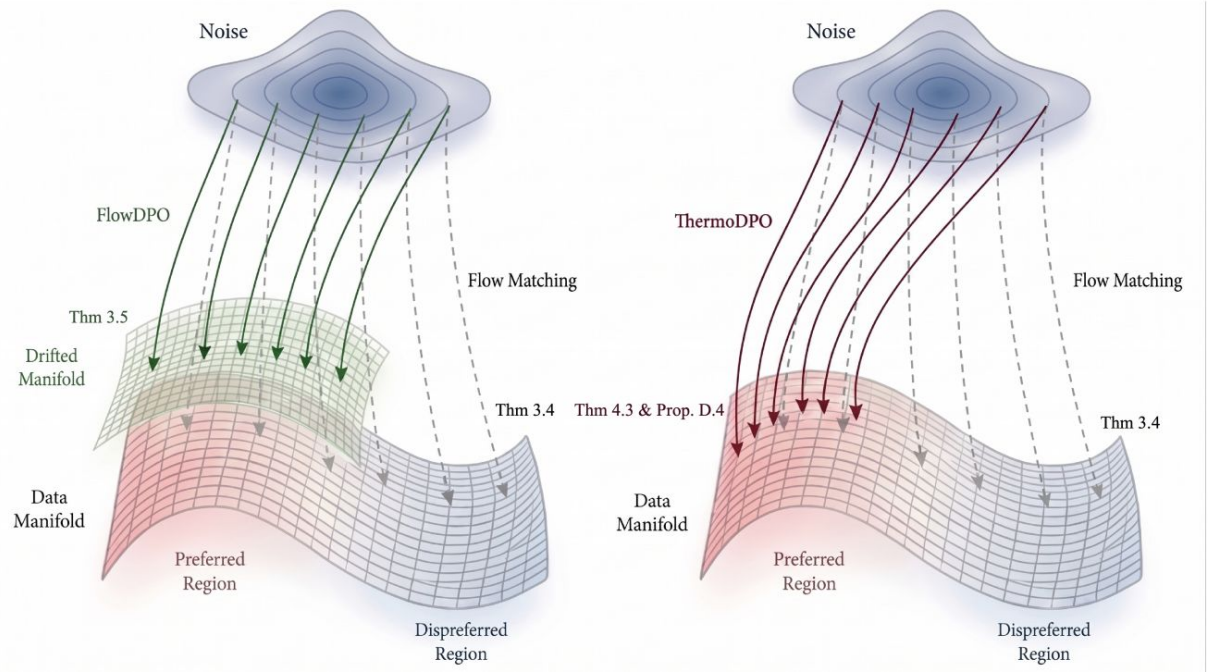
$$\mathcal{L}_{\mathrm{BDPO}}(\pi_\theta; \pi_{\mathrm{ref}}) := -\mathbb{E}_{\mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(\mathbf{y}_w|\mathbf{x})}{\pi_{\mathrm{mix}}(\mathbf{y}_l|\mathbf{x})} - \hat{s}_{\mathrm{ref}}(w; l) \right) \right],$$

where $\pi_{\mathrm{mix}}(\mathbf{y}|\mathbf{x}) = \lambda \pi_\theta(\mathbf{y}|\mathbf{x}) + (1-\lambda)\pi_{\mathrm{ref}}(\mathbf{y}|\mathbf{x})$, for $\lambda \in (0, 1)$.

ThermoDPO

- ThermoDPO also balances DPO <-> SFT, but in a different way

ThermoDPO


Noise
FlowDPO
Flow Matching
Thm 3.5
Drifted Manifold
Thm 3.4
Data Manifold
Preferred Region
Dispreferred Region
Noise
ThermoDPO
Flow Matching
Thm 4.3 & Prop. D.4
Thm 3.4
Data Manifold
Preferred Region
Dispreferred Region

ThermoDPO

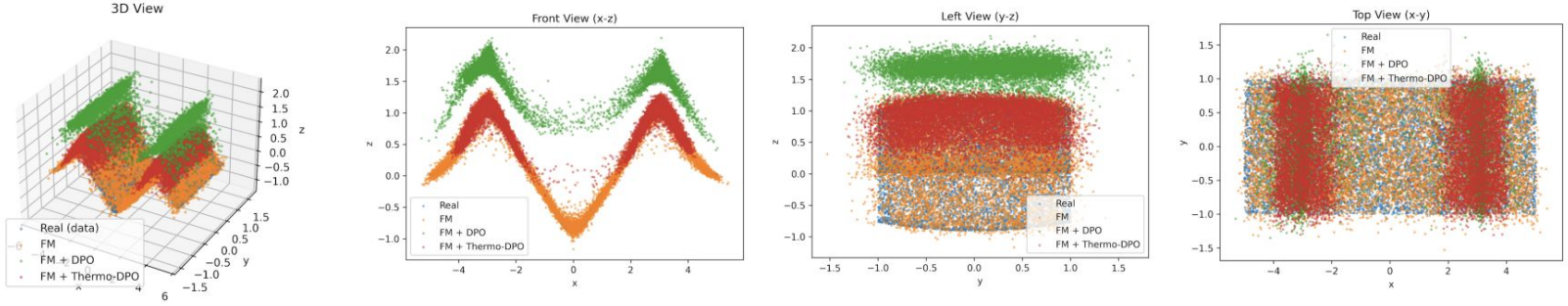


Figure 3: **Toy example of manifold drift under direct preference optimization**. Starting from the same pretrained flow matching (FM) reference model, FlowDPO drives terminal samples toward preferred regions but also causes a deviation from the pretrained data manifold. In contrast, THERMODPO variant preserves the overall manifold structure much better while still improving alignment with the preference signal.

ThermoDPO

Following the FlowDPO notations from Sec. 2, we define three energy-like components that represent the preferred, rejected, and reference states:

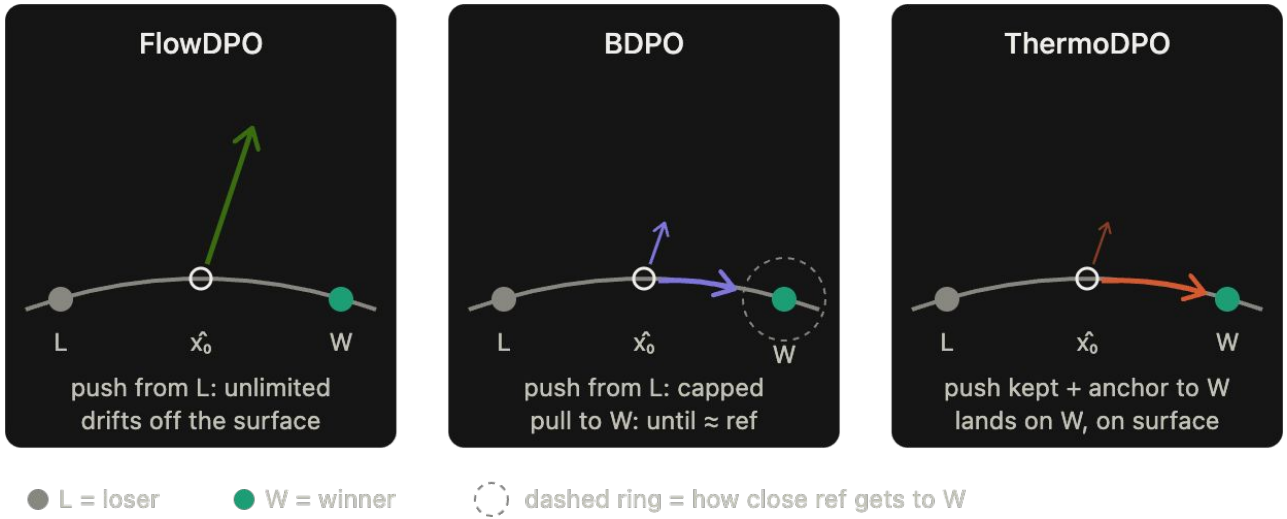
$$E_w = -\frac{\Delta_\theta^w}{\tau(t)}, \qquad E_l = -\frac{\Delta_\theta^l}{\tau(t)}, \qquad E_b = \frac{\ell_{\text{ref}}^w}{\tau(t)}. \tag{4}$$

The THERMODPO objective then minimizes the negative log-probability of the preferred sample within this three-state system, effectively anchoring preference optimization:

$$\mathcal{L}_{\text{ThermoDPO}}(\theta) = \mathbb{E}\left[-\tau(t)\cdot t^2 \cdot \log \frac{e^{E_w}}{e^{E_w} + e^{E_l} + e^{E_b}}\right]. \tag{5}$$

$$\mathcal{L}_{\text{ThermoDPO}}(\theta) = \mathbb{E}\big[g_\tau(\theta)\big], \quad g_\tau(\theta) := t^2 \cdot \tau \cdot \log\left(1 + \exp\left(\frac{\Delta_\theta^w - \Delta_\theta^l}{\tau}\right) + \exp\left(\frac{\ell_\theta^w}{\tau}\right)\right).$$

ThermoDPO vs BDPO



I think doing DPO on x0-JiT models should be much easier

Outcome

- DPO is all about data – Higher margins easier training
- Main feature is off-policy: easy to collect data, but easy to fall into saturation. (even cfg creates biases but you can do cfg distillation implicitly)

References

Direct Preference Optimization (DPO) — https://arxiv.org/abs/2305.18290

Diffusion-DPO — https://arxiv.org/abs/2311.12908

Pick-a-Pic — https://arxiv.org/abs/2305.01569

FiFA: Automated Filtering of Human Feedback Data — https://arxiv.org/abs/2410.10166

Rethinking DPO in Diffusion Models — https://arxiv.org/abs/2505.18736

ThermoDPO · Manifold Drift — https://arxiv.org/abs/2608.20011

Diffusion-SDPO — https://arxiv.org/pdf/2511.03317